

Requirement for SW Developers

- Vending Machine -

Requirements

Implement the central business logic of a vending machine with money change functionality. The vending machine should have multiple slots with an individual price for different drinks. Provide the logic by a reasonably typed interface which allows to verify the functionality by exemplary automated tests. The main functions may e.g. look like this:

```
TotalDepositAmount insertCoin(Coin coin)
DrinkAndChange buyDrink(Drinkrequest selection)
```

The function *buyDrink* either provides the requested drink and potential change in the form of coins with different value or an error, e.g. if the requested drink is out of stock, if the payment is not sufficient, or if the vending machine is not able to change. The vending machine should support the following coin values:

- 10 Cent, 20 Cent, 50 Cent
- 1 Euro (= 100 Cent), 2 Euro (= 200 Cent)

When buying a drink with a price of 1,20 Euro paid by a 2 Euro coin, the vending machine may e.g. provide the 80 Cent change by one 50 Cent, one 20 Cent, and one 10 Cent coin. The drink and money change stocks are limited, and the interface should provide administrative functions for charging and discharging the vending machine.

Consider only obviously required flexibility in your design, e.g. different prices for drink for vending machines in different places, but e.g. no differing coin values. The kata neither demands a user interface nor the persistence of the machine state. A transient solution is sufficient.

The journey is the reward

The objective of this kata is an elegant, simple to understand solution which – as far as possible – makes use of natural terms. Use an object-oriented style in general and the specific capabilities of Java and its standard APIs. Avoid over-engineering and the usage of third-party libraries (except JUnit as test framework).

Minimum Viable Product

The completeness of the solution is of secondary importance. A partial solution is welcome as long as it is consistent, designed in a clean way and validated by automated tests.

Omit the money change functionality for a first increment. Provide an error when the deposit amount exceeds the price.